Spencer Levine

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Mechanical engineering student seeking hands-on experience with the fabrication of robots, complex mechanical assemblies, and part design. Long-term goal: developing robots for lunar resource extraction. Outside of engineering: skiing, playing guitar, political science, space, and I can backflip.

Education

Tufts University School of Engineering Expected Graduation: May, 2028
Bachelor of Science in Mechanical Engineering
GPA: 3.69 / 4, Dean's List
Relevant Coursework: Computer-Aided Design, Materials & Manufacturing I, Mechanics I & II, Engineering Design I, Thermal Fluid Systems I, Ethics of Product Design, Energy Storage Systems

Relevant Work Experience

Nemitz Robotics Group Medford, Massachusetts
Intern Apr 2026 - Present
- Create fixtures and machines to aid in the autonomous assembly of crab robots to disable mines in former war zones, including a motor dispenser and autonomous coupler attachers

John Jay Robotics Cross River, New York
FIRST Robotics Mentor/ Alumni Mar 2024 - Present
- Mentor four robotics teams of high school students competing in the FIRST Tech Challenge (FTC), including one that qualified for the FTC World Championship
- Engage 35+ non-engineering-background students in skills such as CAD, Java coding, and basic robot design
- Co-led high school team to the FTC NY State Championship and the World Championship

UTX Inc. Holmes, New York
Intern May 2026 - Aug 2026
- Manufactured high-tolerance transducers that are used to inspect the integrity of materials
- Designed and created fixtures that increased the speed of inspection and production of transducers by 15%
- Converted 40+ 2D drawings into 3D models; also taught two employees with limited experience to use Fusion
- Programmed plugins for Fusion that create knurls and streamline the modeling of transducer housings

Elaine Short Human Robot Interaction Lab Medford, Massachusetts
Research Assistant Jan 2026 - April 2026
- Designed a low-cost robotic gripper using 3D printing, 25 times cheaper than commercially available grippers
- Supported PhD research by designing components for robotic environments and model robotic arms

Giuliante Machine Tool Inc Peekskill, New York
Intern May 2025 - Aug 2025
- Modeled and created operation sheets for 20+ parts in SolidWorks to be later machined and sold to 10+ clients
- Operated various machines, including CNCs, shapers, and bandsaws, to make gears and shafts

Prominent Personal Projects

Robotic Arm Jul 2025 - Aug 2025
- Designed and built a robotic arm to explore different mechanical linkages, such as 4-bars and chain-driven components in order to learn how these different mechanisms affect motion and mechanical design tradeoffs
- Built a full CAD assembly of the robot in Fusion, then 3D printed and assembled

Swerve Drive Robot Feb 2025 - Apr 2025
- Built a fully 3D-printed swerve-drive robot with custom gear mechanisms as a challenge for myself to learn more about 3D printing and its capabilities, including the use of TPU filament, and to learn about compound gears
- Fabricated a full CAD assembly in Fusion, coded the robot in Arduino and Python, integrating wireless game controller teleoperation

Skills

CAD: Fusion, SolidWorks, Onshape, AutoCAD, 3ds Max, Revit
Computer: Java, Python, C++, JavaScript, AI Automation/Prompt Engineering, LLM Scripting
Manufacturing: Lathe, Mill, 3D printing, Oscilloscope, Pulser-Receiver